CHANGELOOP

SANKALP 2026 · STUDENTS TRACK · CLIMATE AND WATER

In a zero-discharge dyehouse, the evaporator's fuel bill is set by how much salt goes into the dye bath — not by how much water comes out.

A scheduling choice this morning sets how much fuel the evaporator burns tonight. India already recycles its industrial water under Zero Liquid Discharge. Nobody has costed the energy it takes to keep doing so — and that cost is decided upstream, by a planner who never sees the evaporator.

130 million L/day

ALREADY RECYCLED IN ONE CLUSTER

+25 to 30%

WHAT THAT COMPLIANCE COSTS

700 units

SHUT BY COURT ORDER IN 2011

ON ONE PAGE

ChangeLoop in six boxes

Resource decision intelligence for Indian textile clusters operating under a court-mandated Zero Liquid Discharge regime.

PROBLEM

Compliance is solved. Its energy bill is not.

ZLD recycles the cluster's water and raised operating costs 25 to 30 percent. Pollution-board data puts the charge at ₹150–220 per kilolitre against ₹45 for river water — and ₹375–450 at under-used plants.

INSIGHT

In a closed loop, salt is water.

Reject volume equals salt mass over the RO ceiling, so evaporator steam tracks salt, not water. Cut water 20 percent at constant salt and the energy moves 0.0 percent.

SOLUTION

Price the evaporator into the schedule.

Two human-gated decisions: which plan to run, and when wash-off is finished. 480 candidates scored on water, salt, steam, carbon, cost and ship dates at once.

INNOVATION

Two levers that pull apart.

Counter-current rinsing cuts water but not salt. Low-electrolyte chemistry cuts salt but not water. A water-only tool picks the wrong one with full confidence.

IMPACT

The water lever moves the envelope by nothing.

Doing nothing runs at 132.3 percent of a real machine-shift allowance. Counter-current rinsing runs at 132.3 percent too — identical. Only the salt lever gets inside, at 82.1 percent.

EVIDENCE

Checked against published plant data.

Fed the 18,340 mg/L inlet CPCB measured at a Tirupur unit, the model predicts 30.6 percent reject — inside the 20–30 operators report. 21 of 25 coefficients sourced, zero measured.

The ask: one dyeing unit or effluent plant willing to share two instrument readings — steam flow to the evaporator, and reject conductivity. Everything else is already sourced from CPCB, the CEA, TNERC and pollution-board data. Those two are all that separate a validated model from a measured one.

THE PROBLEM

A seasonal river and a bill nobody costed

The Noyyal has no dry-weather flow to dilute effluent, so discharge was ruled out entirely and Zero Liquid Discharge was imposed on the cluster.

Farmers lost the water first

The litigation was brought by the Noyyal River Ayacutdars Association — a farmers' body. The Orathupalayam dam, built to irrigate about 20,000 hectares, became a repository of polluted water.

Compliance became survival

Zero Liquid Discharge now recycles most of the cluster's water. It also raised unit operating costs 25 to 30 percent, in a cluster dominated by MSMEs that cannot absorb it.

The gap is between systems

The dyehouse ERP plans lots. The treatment plant treats whatever arrives. The regulator monitors the outfall. None of them talk, and often they are different companies.

Reported figures. Sources listed in the appendix and in the product's own claim register.

WHERE THE REAL COST SITS

Recovery is solved. Paying for it is not

Recycled water costs two to five times the freshwater it replaces, because somebody had to boil the reject dry to make it.

WHAT IT COSTS	FRESH	RECYCLED
Water, per kilolitre	₹30 to 60	₹120 to 150
Operating cost impact of ZLD	—	+25 to 30%
Cost of dyed fabric	—	+12 to 15%
With salt and water recovery	—	**about +5%**

The industry already names the lever

The reported route from a 12 to 15 percent fabric cost penalty down to about 5 percent is salt and water recovery.

So the question is not whether salt matters. It is who decides how much salt enters the water in the first place — and nobody is doing that.

Avoiding a litre of demand is worth more than recovering a litre of effluent. That is why ChangeLoop acts upstream.

THE INSIGHT

In a zero-liquid-discharge plant, salt is water

Salt is conserved. Reverse osmosis can only concentrate it to a ceiling before scaling stops it. So by mass conservation:

```
V_reject = M_salt / C_reject_max
```

Reject volume — and therefore evaporator steam, fuel and CO_2 — is set by SALT MASS, not by water volume.

And where it stops

Below a crossover the membrane hydraulic floor takes over and cutting salt buys nothing further. A product claiming that cutting salt always wins would be wrong, so the engine reports which constraint binds on every stream.

COMPUTED LIVE IN THE PRODUCT

Cut water 20%, salt unchanged **0.0%**



evaporator energy does not move

Cut salt 20%, water unchanged **-20.0%**



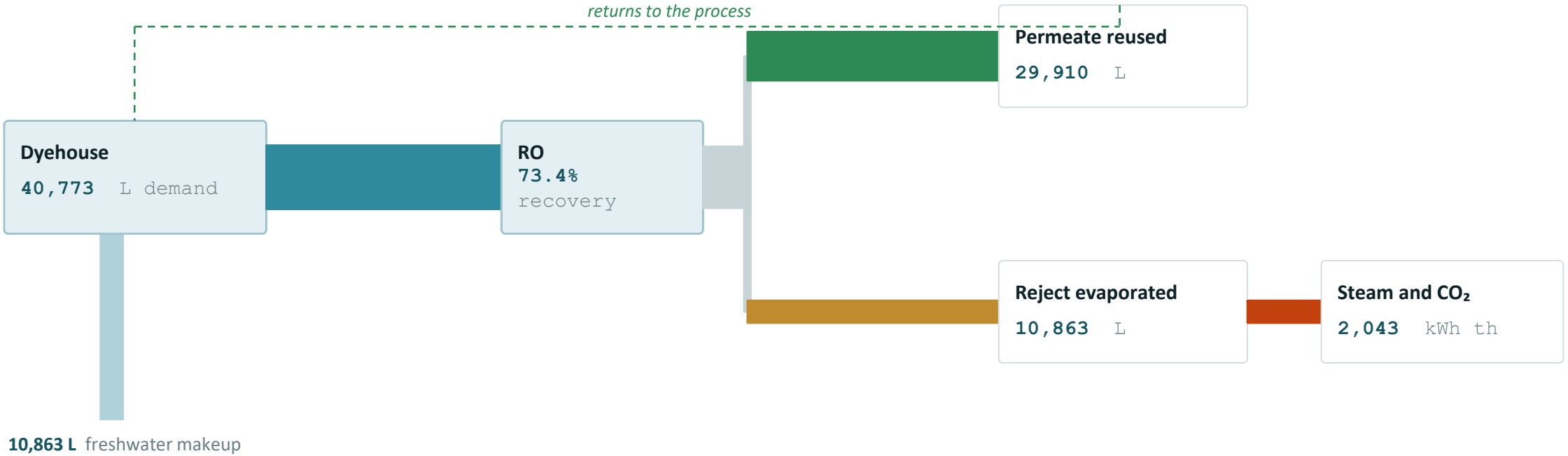
evaporator energy falls proportionally

Cutting water without cutting salt does not cut evaporator energy. Because the loop is closed, freshwater makeup equals exactly what was evaporated — so freshwater, evaporator load and carbon are the same number.

THE SOLUTION

Tonight’s evaporator bill, priced this morning

A decision layer, not a dashboard. It scores every feasible plan on the whole downstream consequence.



Band thickness is volume. The loop that returns is fat; the freshwater top-up is thin — and it is exactly as thick as what the evaporator destroyed.

Two decisions, each gated by a named human

01 **Which plan to run.** Lot order plus process strategy, scored on every consequence at once.

02 **When wash-off is finished.** Releasing early on unfixed dye forces a re-process that costs more than it saved.

THE INNOVATION

Two levers that pull in opposite directions

This is why a joint objective is required, and why a water-only tool picks the wrong lever with full confidence.

PROCESS LEVER	CUTS WATER	CUTS SALT	CUTS EVAPORATOR ENERGY
Counter-current rinse cascade	yes, ~35%	no	barely — same salt, less water
Low-electrolyte dye chemistry	no	yes, ~45%	yes, proportionally

What we claim, narrowly

01 **The downstream consequence inside the objective.** Evaporator steam, boiler carbon and treatment cost, scored per candidate plan.

02 **Salt-mass accounting as the energy driver.** Acting on it inverts the normal advice.

03 **Hard constraints as feasibility, not penalty.** A saving cannot buy its way out of a firm ship date.

What we explicitly do not claim

Ascending shade sequencing is standard dyehouse practice. Counter-current rinsing is established best available technique. Low-electrolyte reactive dyes are a commercial product. Zero Liquid Discharge, reverse osmosis and multi-effect evaporation are all existing industry.
None of these is our invention, and the product says so on screen.

THE PRODUCT, RUNNING

It refuses the plan that saves the most water

480 candidates enumerated exhaustively — 120 lot orders against 4 process strategies — so the optimum is proven, not approximated.

PLAN	FRESHWATER	SALT	EVAPORATOR	CO_2e	COST	LATE
A No intervention	11,907 L	714 kg	2,239 kWh	3,063 kg	₹33,897	0.0 h
B ChangeLoop, recommended	**11,187 L**	**671 kg**	**2,104 kWh**	**2,305 kg**	**₹26,038**	**0.0 h**
C Freshwater minimum — REFUSED	5,889 L	353 kg	1,108 kWh	2,194 kg	₹32,781	2.6 h

Why C is refused

Option C saves almost half the freshwater and ChangeLoop will not recommend it, because it breaches a firm buyer ship date by 2.6 hours.

That refusal is the product working. A water tool that misses a shipment gets switched off in a week.

Every number on this slide is produced by the engine and traceable

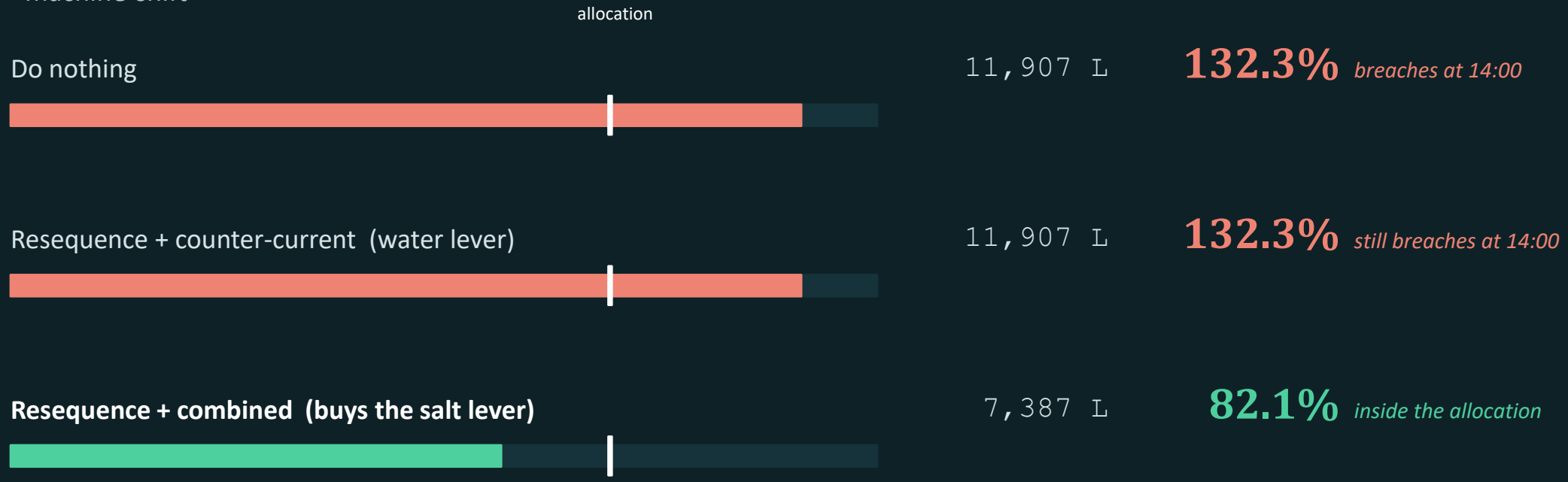
Click any figure in the interface and it opens its formula, its upstream chain, and every coefficient with that coefficient's unit, derivation and evidence class.

8 accounting invariants run on every read of the ledger — water balance, salt balance, no double counting, no credit for an unapproved decision. If one fails, the figure is not displayed.

THE OPERATIONAL MANDATE

The water lever alone does not get you compliant

Judged against a real per-machine-shift abstraction allocation: 72,000 L/day site allowance / (4 machines x 2 shifts) = 9,000 L per machine-shift



Resequencing and reusing rinse water cuts 720 litres and still misses the allocation by a quarter. Only cutting the salt gets the shift inside it. This is the thesis as an operating instruction — and it fell out of the model, not out of a slide.

THE IMPACT

Modest, internally consistent, and traceable

One shift, one machine, five lots. Everything here is MODELLED — nothing in the prototype is metered, and a test enforces that.

1,044 L

FRESHWATER AVOIDED

11,570 L-eq stress weighted, in a basin with no dry-weather flow

62.7 kg

SALT LOAD AVOIDED

this is what sets the evaporator duty

196 kWh

EVAPORATOR STEAM AVOIDED

thermal, at the common treatment plant

926 kg

CO_2E AVOIDED, DOWN 30.2%

₹9,709 saved per shift, down 28.6%

At cluster scale

75.0 million litres of freshwater, **4,500 tonnes** of salt, **14,106 MWh** of evaporator steam and **6,376 tonnes CO_2e** a year, across 400 units on one common effluent plant.

And this is a projection, not a result

It assumes every unit resembles the modelled one and that every recommendation is approved. Neither has been tested. The interface labels it PROJECTED and we will not quote it as achieved impact.

PROOF, NOT ASSERTION

We remove our own layers and publish what breaks

Reproducible from the product's own API. At today's prices, the layers that bind are the salt model and the ship-date rules. Price water scarcity and a scheduler that optimises dyehouse water alone captures 13 percent of the available benefit.

VARIANT	FRESHWATER REPORTED	VS FULL	BREACH	VERDICT
Full ChangeLoop	**720 L**	**100.0%**	**undefined**	**baseline for comparison**
Without downstream ZLD coupling	720 L	100.0%	undefined	no effect at this site
Without the salt-mass model (water-only ZLD)	2,343 L	325.4%	undefined	loses most of the benefit
Without basin stress weighting	720 L	100.0%	undefined	no effect at this site
Without hard-constraint enforcement	1,200 L	166.7%	undefined	loses most of the benefit

Two variants report MORE than is really there

Over-reporting is their defect, not an advantage. A water-only ZLD model mis-states how much freshwater a plan really avoids. We label it as such in the product.

And one of our layers earns nothing here

Basin stress weighting does not change the answer at a single site. It only earns its place for multi-site prioritisation, and our own study says so rather than pretending otherwise.

The cluster is the unit. The CETP is the channel

ROLE	WHY THEY ENGAGE
Unit owner	Buys it. Steam and salt are cash to an MSME.
Production planner	Uses it. Will not accept a missed ship date.
Quality supervisor	Holds the veto on wash-off release.
Common effluent plant	The channel. One integration reaches hundreds of units, and a lower inlet salt load is its own gain.
Buyer or brand	Pulls demand, for Scope 3 water and carbon evidence.

Priced per machine per year, plus one-off integration

Machines, not sites: it scales with the decisions the product actually touches. The business-case screen refuses to compute a return until the site enters its own tariffs — a projection built on a vendor's assumptions is not a business case.

Six-phase pilot, designed to answer one question

Did ChangeLoop cause the reduction, or would it have happened anyway?

1	**Baseline metering** measure before we influence anything	wk 1–4
2	**Shadow mode** we recommend, nobody acts	wk 5–8
3	**Advisory with approval** alternate weeks on and off	wk 9–16
4	**Wash-off release trial** every released lot fastness-tested	wk 17–24
5	**Chemistry decision** once steam cost is measured	wk 25–32
6	**Cluster rollout** second and third unit	wk 33+

A single wash-fastness failure stops the trial. Until phase 3 completes, every figure we show is MODELLED and we will not call it a result.

NOT A MOCK-UP

A working system you can run and attack

Python standard library only. No dependencies, no build step, no account. One command and the whole decision loop is live.

01
Observe
order book and basin allocation

02
Project
causal forecast vs the envelope

03
Optimise
480 candidates, optimum proven

04
Decide
a named human approves or refuses

05
Gate
fail-closed wash-off release

06
Account
mass balance and 8 invariants

07
Prove
tamper-evident ledger and export

190
AUTOMATED TESTS
engine, published-data validation, concurrency, safety and tamper

34
API ENDPOINTS
every figure on screen is server-computed and traceable

8
OPERATING SCREENS
command, forecast, decisions, water, impact, evidence, audit

5
FAULT MODES TESTED
each must force lockout and credit exactly zero

0
MEASURED COEFFICIENTS
enforced by a test, because nothing here is metered yet

Attack it live: inject a probe fault and release is refused at the gate and again at the API, the refusal is written to the ledger, and the credited saving stays at zero. Edit a ledger row by hand and the chain names exactly which record broke.

HOW WE CHECKED A MODEL WITH NO METERS

Fed real plant data, it predicts what real plants report

We have measured nothing, and we say so on every screen. But a model with no measurements can still be falsified — by asking whether its OUTPUTS land where real plants actually sit.

FACT ONE · MEASURED BY CPCB

18,340 mg/L of dissolved solids entering the evaporator at an assessed Tirupur dyeing unit.

FACT TWO · REPORTED BY OPERATORS

RO reject runs at 20 to 30 percent of inlet volume across Indian textile ZLD plants.

Different sources. Neither one is an input to our engine. So we feed it the first and check whether it predicts the second.

INLET TDS FED IN	WHAT THAT NUMBER IS	MODEL PREDICTS	VERDICT
12,000 mg/L	Lower end of observed CETP inlet strength	20%	inside 20–30%
15,000 mg/L	Midpoint of observed CETP inlet strength	25%	inside 20–30%
18,340 mg/L	**TDS measured by CPCB entering evaporation at an assessed Tirupur unit**	**30.6%**	**inside 20–30%**

Nothing is tuned to make this work. The only inputs are conservation of salt mass and the 60,000 mg/L concentration ceiling. The published band is reproduced from floor to ceiling, and a test fails the build if it stops being.

This is cross-validation against published operating data. It is not measurement, and we never call it that.

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COEFFICIENTS SOURCED

CPCB, CEA, TNERC, pollution board, EU BAT

0

MEASURED

and we will not pretend otherwise to win a prize

THE TEAM

Built, tested and adversarially reviewed

Shorya Ashish Mittal

TEAM LEADER

Engine, water accounting and interface

What exists today

A working engine and interface, dependency-free. **190 automated tests** across 7 suites: the engine, cross-validation against published plant data, registry concurrency, safety interlocks over five fault modes, a determinism check, and a tamper test that breaks the ledger chain on purpose to prove it detects it.

Evidence discipline: 17 published, 4 derived, 4 assumed and **zero measured** coefficients — enforced by a test.

Our biggest weakness, stated first

We have not yet spoken to a dyehouse. Everything is modelled from public information, the basin weighting is a disclosed placeholder, and the optimiser handles one machine rather than a mill. Phase 1 of the pilot exists to replace our assumptions with their measurements — and some of our conclusions will change when it does.

What we are asking for: one Tirupur dyeing unit or CETP willing to share two instrument readings: steam flow to the evaporator, and reject conductivity. Everything else is already sourced from CPCB, the CEA, TNERC and pollution-board data — those two are all that separate a validated model from a measured one.

Zero Liquid Discharge gave India back its water. Nobody costed the fuel it takes to keep it — and that cost is decided by a scheduling choice nobody connects to it.